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Using High-Probability Request Sequences to Increase Social Interactions in Young Children With Autism

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We investigated the effects of an intervention using high-probability request sequences with embedded peer modeling to increase social interactions of children with autism in a classroom. The effects of the intervention on compliant responding to social requests and social behaviors were monitored using a single-subject multiple baseline design across children. Additionally, social validity regarding the intervention goals, procedures, and outcomes was measured by relevant consumers. The results of this study indicate that all three children's compliant responding to low-probability requests and social behaviors increased with the intervention and were maintained. Furthermore, the target children's social behaviors generalized to untrained peers and nontraining settings. The social validity results indicated a high level of consumer acceptability and usability among relevant consumers.

Keywords: *autism; high-probability requests; social interactions; social validity; peer-mediated interventions*

The lack of social skills is a distinctive characteristic of children with autism (American Psychiatric Association, 2000). Researchers assessing social behavior note that children with autism show limited interest in peer play and rarely interact with their peers (e.g., Koegel, Koegel, Frea, & Fredeen, 2001). Conversely, children without developmental delays engage frequently in social interactions with peers. With substantial social skill deficits, children with autism may have difficulty developing positive peer relationships in inclusive environments (DiSalvo & Oswald, 2002; McConnell, 2002; Strain & Hoyson, 2000).

A number of intervention strategies have been documented to increase the social interactions of young children with autism (Brown, Odom, & Conroy, 2001; Kohler, Anthony, Steighner, & Hoyson, 2001; Odom et al., 1999; Tsao & Odom, 2006). Among these strategies,

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peer-mediated ones are the most frequently cited and yield the strongest empirical support for promoting social behavior (Brown & Odom, 1995; Kohler et al., 2001; Rogers, 2000). Peer-mediated interventions focus on training and prompting peers to interact with target children. Whereas peer-mediated interventions promote social interactions between children with autism and their peers, children with autism are primarily engaged in responding to their peers' initiations rather than initiating social behavior. This may explain, in part, the observation by researchers that peer-mediated interventions often resulted in limited generalization and maintenance of improved social interactions (Chandler, Lubeck, & Fowler, 1992; Krantz, 2000; Strain & Kohler, 1998).

Whereas generalization of treatment effects across settings and time remains a continuing issue in social skill intervention strategies (McConnell, 2002; Odom et al., 1999), researchers have suggested that interventions focused on promoting certain social behaviors (e.g., sharing, complimenting, affectionate behaviors) will have a high probability of receiving positive responses from peers and facilitate positive social interactions (Brown & Odom, 1994; Fox & McEvoy, 1993; Twardosz, Nordquist, Simon, & Botkin, 1983). In addition, teaching multiple social behaviors in various settings (Brown & Odom, 1994, 1995) and providing the intervention in the context of ongoing activities and routines (Kohler, Strain, Hoyson, & Jamieson, 1997; Rogers, 2000; Strain & Schwartz, 2001) have also shown to facilitate generalization.

Another promising strategy for both the facilitation and generalization of social interactions of children with autism is the use of high-probability (hereafter referred to as *high-p*) request sequences (Davis, Brady, Hamilton, McEvoy, & Williams, 1994). The high-p intervention includes the delivery of a sequence of requests that a child is more likely to perform (i.e., a high-p request) presented immediately before the delivery of a request not typically performed (i.e., a low-probability request; hereafter, *low-p* request) (Davis & Brady, 1993). The high-p request procedure is implemented as follows: (a) Deliver a sequence of three to five high-p requests at a rapid pace, (b) deliver praise for each performance of the high-p requests, (c) deliver a low-p request, and (d) deliver praise for the performance of the low-p request (Davis, 1995). This tactic has been successful in facilitating compliant responding to general requests in a variety of settings (Ardoin, Martens, & Wolfe, 1999; Davis et al., 1994; Davis, Brady, Williams, & Hamilton, 1992; Davis, Reichle, & Southard, 2000; Killu, Sainato, Davis, Ospelt, & Paul, 1998; Mace et al., 1988; McComas, Wacker, & Cooper, 1998).

Several investigators have extended the use of high-p request sequences beyond the promotion of compliance to general instructions by using the intervention tactic to promote communication and social skills in young children with disabilities (Davis et al., 1994; Davis & Reichle, 1996; Sanchez-Fort, Brady, & Davis, 1995; Santos & Lignugaris/Kraft, 1999). In these studies, the high-p request procedure was combined with other intervention strategies. For example, to improve communication skills, Sanchez-Fort et al. (1995) and Santos and Lignugaris/Kraft (1999) combined a high-p request procedure with a milieu language intervention strategy. In another study, Davis et al. (1994) demonstrated that the high-p intervention combined with generalization strategies increased social interactions of young children with autism. These effects were maintained and generalized to untrained peers and in nonintervention settings. In a subsequent study, Davis and Reichle (1996) demonstrated that high-p request sequences combined with peer-mediated strategies and delivered in a variant sequence by peers increased compliance with social requests in young children with emotional–behavioral

disorders. Although no data on unprompted social behaviors after removing the high-p intervention were collected in this study, results indicated that varying the sequences of the requests was critical to the effectiveness of the intervention strategy. In summary, the high-p request intervention has demonstrated its utility in teaching a variety of functional skills embedded into existing classroom routines using typical instructional strategies (e.g., modeling or reinforcement). To enhance the effect of the high-p intervention on unprompted social behaviors, it may be necessary to embed components that occur naturally in the child's classroom environment, such as the use of peers as intervention agents.

Young peers provide many opportunities for facilitating social interactions in ways that adults cannot (Erwin & Schreiber, 1999). Peers may act as models for language and social behavior. Investigators have noted that the use of peers may enhance positive effects on children's social behaviors and generalized imitation within classroom routines (e.g., Garfinkle & Schwartz, 2002; Jones & Schwartz, 2004; Kohler & Strain, 1999; Werts, Caldwell, & Wolery, 1996). Embedding peer modeling in a social skill intervention may provide children with autism more opportunities to observe peers' extended play within the play context as well as to initiate and respond to their peers as play partners. Combining a peer modeling procedure with the high-p request sequence may increase generalized social outcomes through providing social learning opportunities for children with autism.

To develop social skill interventions that are useful to practitioners, the feasibility, acceptability, and effectiveness of the intervention should be considered as important factors to be measured (McConnell, 2002; Strain & Schwartz, 2001). Most social skill intervention strategies require extensive teacher involvement to implement with typically developing peers and children with autism (Kohler et al., 2001). Although researchers employing high-p request interventions have demonstrated that the procedure can be incorporated into classroom routines or existing intervention programs in preschools without extra materials (Davis, 1995; Davis et al., 1992, 1994, 2000; Davis & Brady, 1993; Davis & Reichle, 1996; Killu et al., 1998; Sanchez-Fort et al., 1995; Santos & Lignugaris/Kraft, 1999), few investigators have reported social validity measures of the high-p intervention (Ardoin et al., 1999; Davis et al., 2000; Ducharme & Worling, 1994; Killu et al., 1998).

Whereas a number of social skill interventions have been developed to improve the social behaviors of young children with autism, a valuable addition may combine the effective components of empirically validated strategies that are developmentally appropriate and can be incorporated into common classroom activities. Strong empirical support exists for the use of peers to teach targeted social behaviors to children with autism. Although numerous investigators have demonstrated the effectiveness of the high-p request sequence, few have been conducted using this tactic with young children and to increase the social interactions of young children with autism (Davis et al., 1994). To date and to our knowledge, no social skill researchers have combined the high-p intervention with embedded peer modeling in classroom contexts.

The purpose of our study was to examine the effects of high-p request sequences with embedded peer modeling on the compliant responding to social requests in young children with autism. We attempted to increase the social interactions of children with autism with their peers without developmental delays in the intervention and generalization settings. Furthermore, we investigated the effects of the intervention on relevant consumers' assessments of the intervention goals, procedures, and outcomes.

Method

Participants

This study was conducted in an inclusive private charter school in a large midwestern metropolitan area. Three children with autism spectrum disorders participated in the investigation along with 6 children without developmental delays who were enrolled in the same class and served as peer partners. The participants' cognitive and language skills and autistic characteristics were assessed using the Battelle Developmental Inventory (Newborg, Stock, Wnek, Guidubaldi, & Svinicki, 1984) and the Childhood Autism Rating Scale (CARS; Schopler, Recichler, & Renner, 1988).

All 3 participants demonstrated significant developmental delays in cognitive and language skills on standardized tests and scored in the range of mild to severe autism based on the CARS (see Table 1). Each participant had Individualized Education Program (IEP) objectives related to social skills. They were able to perform simple tasks and respond to the teacher with one or two words when requested to do so. During center time, all 3 participants were more likely to remain alone in a corner engaging in stereotypic behavior rather than interacting with toys or peers.

Eric was a 5-year-3-month-old boy with an independent diagnosis of pervasive developmental disorder, not otherwise specified (PDD-NOS). He was able to verbally respond to adults and peers. However, Eric often wandered around the classroom, ignored other children, or exhibited a negative response toward them if they attempted to engage him and often engaged in stereotypic behaviors with toys during play or center time. Paul was a 6-year-3-month-old boy who had an independent diagnosis of PDD-NOS. He was able to respond to teachers' simple and routine requests; however, he seldom interacted with other children and was more likely to avoid other children when they initiated to him. He exhibited disruptive behaviors (e.g., kicking, crying, hitting himself, throwing away task materials) when he was asked to do demanding tasks or requests. Paul often engaged in stereotypic play with several specific toys (e.g., pounding horses, lining up dinosaurs). Chris was a 6-year-5-month-old boy who had an independent diagnosis of autism. He seldom engaged in play and often stared into space while sucking his thumb. When other children attempted to play with him, he frequently yelled or pinched them. Chris often screamed, hit, pinched, or scratched others when he was asked to do demanding tasks or requests.

Six children without developmental delays in the same classroom served as peer partners in the study. Three peers without developmental delays who followed teacher directions consistently, demonstrated age-appropriate interactions with peers, and attended school regularly were selected to receive training and participate in the intervention (referred to as *trained peers*). The remaining 3 peers without developmental delays (referred to as *generalization peers*) were present during the delivery of intervention and the generalization settings but did not participate directly in the intervention. The trained peers were 3 girls ages 4 years 3 months, 4 years 4 months, and 5 years 1 month, respectively. The other, untrained generalization peers were 2 boys and 1 girl ages 5 years 6 months, 4 years 3 months, and 4 years 6 months, respectively.

Table 1
Characteristics of Target Children

Characteristic	Eric	Paul	Chris
Age	5 years 3 months	6 years 3 months	6 years 5 months
Gender	Male	Male	Male
Diagnosis	PDD-NOS	PDD-NOS	Autism
CARS	31.5	33. 5	42
BDI			
Cognition	43 months	35 months	30 months
Receptive Language	32 months	23 months	23 months
Expressive Language	30 months	18 months	16 months

Note: PPD-NOS = pervasive developmental disorder, not otherwise specified; CARS = Childhood Autism Rating Scale (Schopler, Recichler, & Renner, 1988); BDI = Battelle Developmental Inventory (Newborg, Stock, Wnek, Guidubaldi, & Svinicki, 1984). Age-equivalency scores reported in months.

Settings and Materials

This study was conducted in an inclusive preschool–kindergarten transition classroom during center time. The classroom consisted of 6 children with developmental delays and 7 peers without developmental delays as well as a lead teacher and two assistants. The classroom (6.4 m × 5.7 m) was separated into defined play areas, including a kitchen area, art–manipulative area, blocks and trucks area, books–language area, train table, workbench, and a sensory area. Three intervention settings and one generalization setting were selected for each target child depending on the child's preferences and the opportunities for interactive play between the children. To identify child preferences for play areas, the children were observed by the experimenter and teachers during center time activities. In addition, parents were surveyed and asked to identify their children's preferences.

The intervention settings selected for Eric and Paul were blocks and trucks, kitchen area, and train table. The kitchen area, train table, and art–manipulative area served as the activity areas for Chris. For each student, the intervention settings were randomly rotated throughout the study. The sensory area, workbench, and blocks and trucks were selected as generalization settings for Eric, Paul, and Chris, respectively. Each of the intervention and generalization settings was clearly defined by shelving and furniture, and each contained easily accessible age-appropriate toys and materials. During the center time, the teacher assigned small groups of children to the centers. Children rotated with their group between the centers during this time. As the intervention was embedded during the center time, the teacher was asked to assign the experimental groups to the appropriate intervention activity setting or generalization setting.

Behavioral Measures

We focused on facilitating social interactions of children with autism by increasing their responses to social requests. The dependent variables were responses to low-p requests and social interactions. Additionally, disruptive behaviors were recorded. Responses to high-p requests were measured for procedural fidelity.

Responses to low-p requests. During the study, low-p requests were the requests to initiate to a peer, to which the student had a history of being unlikely to respond (i.e., with a criterion of 50% or less). A response to a low-p request was defined as the child's initiating the requested response within 5 s and the successful completion of the targeted low-p request.

Social interactions. Social interactions between the participants and peers without developmental delays were measured. Social interactions were categorized into unprompted and prompted (i.e., response to a low-p request) initiations and subsequent responses (Odom, Chandler, Ostrosky, McConnell, & Reaney, 1992; Odom, Strain, Karger, & Smith, 1987). An unprompted initiation was any verbal or gestural behavior (e.g., touching a peer on the shoulder) that the target child directed to a peer that was not preceded by a social behavior from the same peer in the previous 3 s or the delivery of a high-p request. A response was any verbal or gestural behavior that was directed toward a peer initiation that occurred in the previous 3 s (Odom et al., 1987, 1992). Additionally, social interactions were recorded as to whether they were directed to training peers or generalization peers.

Responses to high-p requests. A high-p request is defined as a request that the child responds to 80% of the time or better. The high-p requests consisted of requests to perform preferred action songs the child liked and play skills appropriate within the given context, for example, "hands on your shoulders" or "cut the pizza." A response to a high-p request was defined as the child's initiating the requested response within 5 s and the successful completion of the targeted high-p request as determined.

Disruptive behavior. Disruptive behavior was defined as any inappropriate verbal or gestural behaviors, such as throwing away the toys, hitting oneself, screaming during play, or other inappropriate responses by the child that were not necessary to perform a request.

Data Collection

The children were videotaped during a 1-hr center time using a video camera. While the sessions were videotaped, an audiotape played to provide an audible cue for each interval that could be later used by observers for coding the videotape. The secondary observer monitored the video camera during all sessions to capture all aspects of the center time conditions. The experimenter served as the primary observer and recorded the occurrences of the dependent variables from the videotapes. A secondary observer was an undergraduate student in special education and was trained using videotapes and practicing observations until the primary and the secondary observer reached at least 95% agreement on at least two consecutive sessions.

The children's behaviors were recorded for a 10-min period of each participant's intervention using a 6-s partial interval and event recording scheme. Observed data included (a) prompted initiations (i.e., those initiations preceded by a low-p request), (b) unprompted initiations, (c) unprompted responses to initiations, (d) delivery of and response to high-p and low-p requests, (e) interprompt time (i.e., time between the reinforcement of the previous request and the delivery of the next request), (f) disruptive behavior, and (g) the experimenter's

general prompts. The children's discrete social initiations, responses, and disruptive behaviors were measured with a 6-s partial interval recording. The children's responses to high-p and low-p requests were recorded using event recording. Responses to high-p and low-p requests were converted to the percentage of appropriate responses to requests. The percentage of correct responses to requests and the number of intervals of social behaviors were the basis of the experimental analysis and graphed for visual inspection.

Interobserver Agreement

Interobserver agreement was collected and analyzed to ensure the believability of the coded data. The experimenter and second observer jointly viewed the videotaped sessions and independently recorded the dependent variables and experimenter behaviors. For any disagreements between the experimenter and the second observer, the videotaped session was viewed a second time and corrections were made. Interobserver agreement was measured for the children's responses to the requests for an average of 25% of the experimental sessions and calculated by using an (agreements/agreements + disagreements) × 100 formula. Interobserver agreement for responding to the requests between the two observers was 100% for all participants during the initial low-p, the high-p, and the second low-p conditions. Interobserver agreement for social initiations and social responses was collected on 32% of the experimental conditions and 35% of the generalization sessions. Mean interobserver agreements for social initiations and social responses across the experimental conditions were as follows: Eric, 98%, 98%; Paul, 99%, 99%; and Chris, 99%, 98%, respectively. Mean interobserver agreements for social initiations and responses across the experimental conditions in the generality setting were as follows: Eric, 97%, 97%; Paul, 99%, 99%; and Chris, 99%, 99%, respectively.

Procedural Integrity

Procedural integrity measurements were conducted to ensure that the experimenter accurately and consistently implemented the intervention with each child. Data were collected using an implementation checklist, delivery of low-p and high-p requests, and the experimenter's general prompts. A checklist was used to mark each procedural step as either observed (+), not observed (–), or not applicable (NA). Two undergraduate students majoring in special education completed the procedural integrity checklist during 32% of the experimental sessions and 35% of the generalization sessions. Interobserver agreement for procedural integrity checklists was 100% for participants during all conditions.

Using an event recording scheme, the experimenter's delivery of each low-p and high-p request sequence was recorded by the experimenter and the second observer while viewing videotaped sessions. The observers ensured that at least three high-p requests, correct responses to high-p requests, and reinforcement for performance of high-p requests were delivered prior to the delivery of a low-p request. In addition, the interprompt time (i.e., time between the reinforcement of the previous request and the delivery of the next request) was measured to ensure that the experimenter delivered each request within 5 s after reinforcement for performing the last request. Two observers observed an average of 25% of the experimental sessions during the initial low-p, the high-p, and the second low-p conditions, and the interobserver agreement was 100% between the two observers across all conditions for all participants.

Table 2
The Mean Number of Intervals and Range of General Prompts for Participants During the Intervention and Generalization Settings

	Intervention Settings					Generalization Setting				
Child	BL	Low-P	High-P	Low-P	Maint.	BL	Low-P	High-P	Low-P	Maint.
Eric										
Mean	4.33	4.40	1.54	1.13	1.33	4.50	3.00	1.80	1.75	1.50
Range	2-7	0-7	0-5	0-2	0-3	4-5	2-4	0-3	1-2	1-2
Paul										
Mean	5.10	3.45	1.61	2.18	3.50	6.67	4.00	2.83	2.67	2.50
Range	2-15	1-7	0-3	0-5	2-5	2-4	2-6	2-4	2-4	2-3
Chris										
Mean	5.00	5.80	2.16	2.57	3.00	2.50	4.33	1.80	2.00	–
Range	2-13	2-11	0-4	0-5	3	1-4	3-5	1-3	0-4	–

Note: BL = baseline; low-p = low-probability request; high-p = high-probability request; maint. = maintenance.

In addition, the number of intervals of experimenter's general prompts for stereotypic or disruptive behaviors delivered to each participant was recorded for all conditions. Interobserver agreement was collected during an average of 32% of the experimental sessions and an average of 35% of the generalization sessions. Interobserver agreement between the two observers in the experimental sessions was 99%, 99%, and 99% for Eric, Paul, and Chris, respectively. Interobserver agreement between the two observers in the generality sessions was 100%, 99%, and 99% for Eric, Paul, and Chris, respectively. A summary of the mean number of intervals and range of general prompts is presented in Table 2.

Social Validity

To assess social validity, relevant consumers, including the parents of the target children and professionals in early childhood special education, were asked to provide information regarding the social importance of the intervention goals, procedures, and outcomes. Social validity of the intervention goals was assessed by gathering information from three parents of the children participating in the study and three early childhood special education teachers at the school. Parents and teachers were asked to complete a questionnaire asking whether they considered each research question to be valid and useful.

To determine the social acceptability and usability of the intervention procedures, the experimenter gathered information from 20 professionals in early childhood special education. The teachers were asked to read a description of a high-p request sequences with embedded peer modeling. The professionals viewed three 2-min segments of videotape depicting the target children during baseline, low-p request intervention, and high-p request sequences with embedded peer modeling. The viewers were unaware of the experimental condition depicted on the video clip. Next, the professionals were asked whether they thought the intervention would be easy to use during the daily class routine and whether they would use the intervention in their classroom when considering both time and effort necessary for the intervention.

To assess the social validity of intervention outcomes, the same professionals in early childhood special education above were asked to complete a checklist after viewing videotaped segments of all experimental conditions for each child. The checklist consisted of items asking the teachers whether the children complied with the requests, focused on play, shared toys with peers, and responded to peers. Three segments (i.e., baseline, low-p request intervention, and high-p request sequences with embedded peer modeling) for each child were randomly ordered and shown to the teachers. The teachers were asked to view one videotaped segment and then to complete a checklist. After viewing all the taped segments, the teachers were asked to answer follow-up questions regarding whether they thought this intervention would be effective to increase social interactions between children with autism and typically developing peers.

Experimental Design

A multiple-baseline design across participants was used to assess the effects of high-p request sequences with embedded peer modeling on increasing social interactions between children with autism and their peers during center time. The five experimental conditions were as follows: (a) baseline, (b) low-p requests, (c) high-p request sequences with embedded peer modeling, (d) low-p requests, and (e) maintenance (removal of requests). Generalization probes were conducted in the nontraining setting during center time, which followed the experimental sessions for all conditions. Prior to the experiment, normative data on social interactions of peers without developmental delays in the classroom and the status of the low-p and the high-p requests were assessed.

Procedures

Normative measurement. Prior to baseline, observations were conducted to obtain normative data on the numbers of intervals with social initiations and social responses among peers without developmental delays using a 6-s partial interval recording scheme. Five typically developing children were randomly selected in the same class. The experimenter collected data on the children's social initiations and responses toward other children in a training setting for 5 days. The average numbers of intervals of social initiations and social responses for the 10-min play time during center time were 7.4 and 1.6, respectively. During the 10-min play session, the average number of intervals of total social behaviors (i.e., initiations and responses) was 9. These data provided a guide to determine how frequently a low-p request (i.e., prompts for social initiation) should be issued to approximate natural peer interactions.

Status assessment of high-p and low-p requests. The pool of high-p and low-p requests was determined by (a) observing a child's preference during the center time, (b) surveying the teachers' and the parents' opinions, and (c) examining the children's IEP. Subsequently, a pool of requests appropriate to the children's play skills and performance levels in the settings was determined. In addition, requests were chosen from the children's favorite action songs and group affection activities that were more likely to receive positive responses from other

children (Cooper & McEvoy, 1996). The high-p and low-p status of requests in each intervention setting was determined by assessing each child's response to the pool of requests. Each request was delivered 10 times to each child, and a percentage of correct responses was calculated during a 3-week assessment period. The low-p requests (i.e., requests for social initiations) were those opportunities to which the children responded less than 50% of the time. The high-p requests were those opportunities to which the children responded 80% or more of the time. However, these criteria were adjusted according to the levels of the individual child's responses to the requests. For example, Eric's low-p requests were those to which he responded 60% or less of the time. Chris's high-p requests were those to which he responded 70% or more of the time. Four low-p requests in each of the three intervention settings and 4 requests adapted from group affection activities were determined. The total low-p requests per child were 16 requests (i.e., 12 low-p requests for three intervention settings and 4 low-p requests from friendship activities). Thirty-three high-p requests each for Eric and Paul and 28 high-p requests for Chris were determined.

General procedures. During center time, the classroom routine was to have the children play in the two centers assigned by the teacher. Children could move from center to center within the assigned two centers. The experimenter assigned one intervention center or generalization center, and the teacher assigned the remaining center. The participants were required to play in an intervention or generalization setting. They then moved to the next center assigned by the teacher with the other children. The second observer started the interval tape after the experimenter gave the general instruction to the target child and the peer and cued the experimenter to conclude after 10 min. One 10-min intervention session was conducted per day for each group. Three intervention settings were randomly rotated daily for each participant.

Baseline. After the children were assigned to an intervention center, the target child and his peer were instructed, "Stay in your center and play with your friend. Now go and play." The generalization peer remained in the center but was not given any explicit instruction. The experimenter gave general prompts to play only when the child engaged in stereotypic or disruptive behaviors. The experimenter did not give any specific prompts to interact to the children. After the 10-min play session, the children were verbally praised for participating in the center.

Peer training. In the peer training condition, the experimenter trained the peer to recognize the target child's initiations and to respond to them by using role-play and discussion. All training occurred in the classroom and was specific to the target child's preferred play centers. The second observer or third observer played the role of the target child. The experimenter delivered the low-p requests to the observer and the observer responded to the experimenter's low-p requests as if she were the target child. The peer was told to wait until the target child followed the experimenter's instruction when the experimenter told the target child what to do. The experimenter asked the peer what her response might be to the initiation. When the peer did not respond appropriately, the experimenter modeled the response and told the peer how to respond to the initiation. The same procedures continued for all target behaviors in the three intervention settings for 10 min until the peers reached the criterion of 90% correct responses to the experimenter's requests for three consecutive sessions.

Low-p requests. The setting, context of play, and the experimenter's general prompts remained the same as in the baseline condition. The experimenter delivered eight low-p requests to each target participant in the 10 min center time. Eight low-p requests included four requests adapted from group affection activities (e.g., "Shake hands with your friend," "Say hi to Hilary") and four other requests relevant to the assigned play setting (e.g., "Push the train to Stacie," "Give Jennifer the pizza"). Low-p requests were written on individual index cards to cue the experimenter on the delivery of requests. Before each session, the experimenter shuffled the cards with low-p requests to vary the order of their delivery. However, the low-p requests included group affection activities that were delivered at the beginning or end of the session in the appropriate context.

The experimenter delivered a low-p request to the target child. If the target child performed the request within 5 s, the experimenter provided verbal and gestural praise. If the target child did not perform the request, the waited approximately 75 s, an amount of time that was based on the normative data and previous studies (Davis et al., 1994; Davis & Reichle, 1996). After 75 s, the experimenter delivered a new low-p request. No consequence was provided for nonresponse to requests.

High-p request sequences with embedded peer modeling. Once stable responding was established, the high-p request sequences with embedded peer modeling was implemented. The same eight low-p requests as in the previous low-p request condition were used. High-p requests consisted of requests to perform play skills relevant to the play setting (e.g., "Cut the pizza," "Build an airport") and requests adapted from action songs (e.g., "Hands on your shoulders," "Move your eyebrows"). The high-p request sequences for each child were written on index cards to cue the experimenter and were shuffled every session so that the order of delivery was varied. The high-p request sequence was first delivered to the peer as a model, and then the same high-p request sequence was delivered to the target child. In effect, the target child observed how the peer responded to the requests and how she was praised. Furthermore, the target child had the opportunity to respond to the peer's social initiations by the peer's responding to the high-p sequences as a model.

The high-p request sequence included the following: (a) rapid-paced delivery of three high-p requests (i.e., within 5 s of each other), (b) the delivery of verbal or gestural praise (e.g., tickling, patting, thumbs-up), and (c) the delivery of the low-p request within 5 s after reinforcement for performing the previous high-p request. After issuing the high-p request, if the child did not perform the request, a different high-p request was delivered 5 s after the delivery of the previous request. The experimenter continued to deliver high-p requests until three consecutive responses were obtained. The low-p request was delivered immediately (within 5 s) after reinforcement for performing the last high-p request. If the child did not respond to the low-p request, the experimenter issued the next high-p request sequence.

Low-p requests. During this condition, the high-p request sequences with embedded peer modeling were withdrawn and remained the same as in the previous low-p request procedure. This condition was provided to monitor whether the child's responding to the low-p requests would be maintained after removal of the intervention.

Maintenance. The setting and context of play remained the same as in the baseline phase. The children were told to play with the toys and their peers without the low-p requests. This condition was provided to assess the extent to which the increased social interactions were maintained without the prompts.

Generalization. Generalization probes were conducted in the nontraining setting during center time. Procedures were identical to baseline procedures. After 10 min of play, the children were verbally praised and were told that they could stay or change centers.

Results

Data on the dependent variables across all conditions are presented for compliant responding to low-p requests, social initiations and responses, and disruptive behaviors. In addition, social validity results from parents and the professionals in early childhood special education are presented.

Responses to the Requests

Data on the participants' compliant responding to low-p requests were collected during the initial low-p request condition, the high-p request sequences with embedded peer modeling, and the second low-p request condition. During the baseline and maintenance conditions, no requests were delivered to observe the social behaviors between children without the intervention.

In the initial low-p request condition, the participants' responses to the low-p requests (i.e., to initiate social behaviors) were low, with means of 34.1%, 37.8%, and 14% for Eric, Paul, and Chris, respectively. Figure 1 presents the data on the percentage of correct responses to the low-p and the high-p requests for Eric, Paul, and Chris.

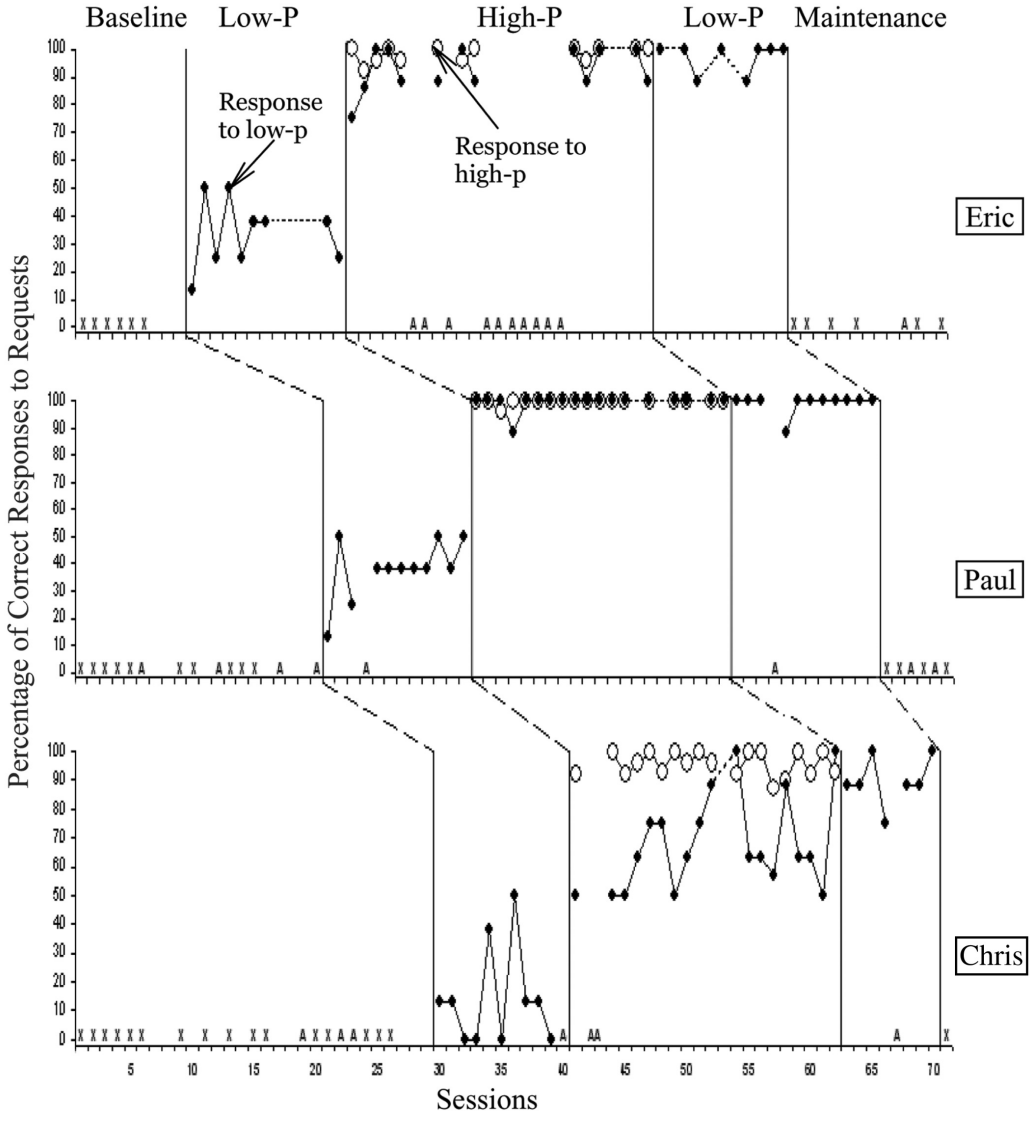
During the high-p request sequences with peer modeling, participants' compliant responses to the low-p requests increased sharply. The mean scores of the percentage of correct responses to low-p requests were as follows: Eric, 92.4%; Paul, 99.3%; and Chris, 67.7%. Additionally, all participants' responses to the high-p requests were high, with means of 98.2 %, 99.6 %, and 95.8 % for Eric, Paul, and Chris, respectively. Trained peers were asked to model the responses to the high-p and the low-p requests. All peer models, Stacie (Eric's peer), Jennifer (Paul's peer), and Hilary (Chris's peer), responded to the low-p requests at 100%. They responded to the high-p requests with means of 98%, 99%, and 100% for Stacie, Jennifer, and Hilary, respectively.

After withdrawing the intervention in the second low-p request condition, the participants' levels of responses to low-p requests remained high. The mean levels of responding to low-p requests were 97%, 98.9%, and 89.6% for Eric, Paul, and Chris, respectively.

Social Interactions in the Intervention Settings

Data on social initiations and responses include the number of intervals of observed social initiations and responses for each child during all experimental conditions and

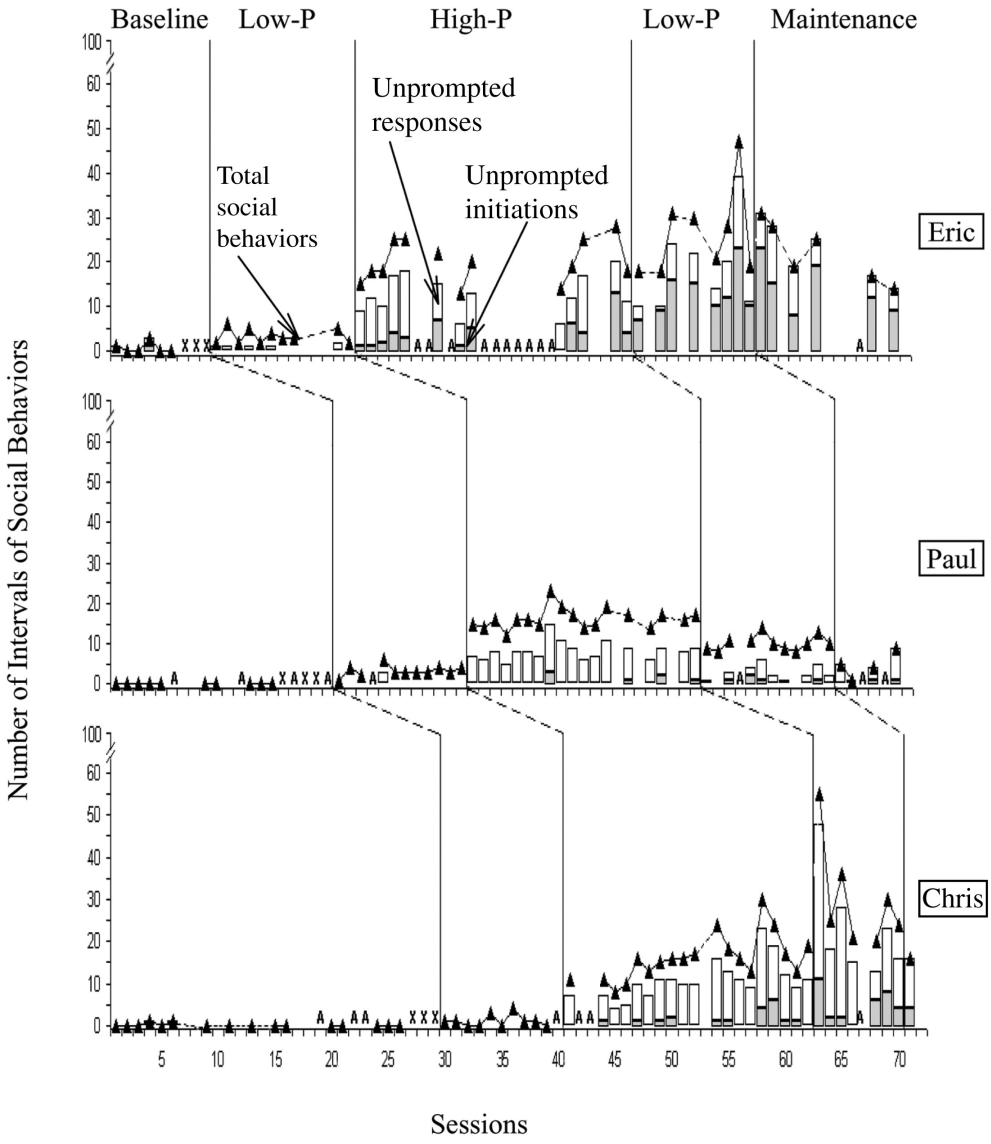
Figure 1
Percentage of Correct Responses to Requests.



Note: Filled circles (-●-) represent percentage of correct responses to low-probability requests, and open circles (-○-) represent percentage of correct responses to high-probability requests; *A* represents absence; *X* depicts the sessions that data were taken without low-probability requests in baseline and maintenance.

generalization sessions. The results for each child's total social behaviors (i.e., the total of prompted social initiations, unprompted initiations, and unprompted responses), prompted social initiations (i.e., responses to the low-p requests), unprompted social initiations, and unprompted responses to the trained peer and the generalization peer are represented for all experimental conditions in the intervention settings. Figure 2 displays the number of

Figure 2
Number of Intervals of Social Behaviors in the Intervention Settings



Note: Bar graphs depict number of intervals of unprompted social initiations and responses to all peers. Data points (▲) represent total number of intervals in which social interactions of both prompted and unprompted initiations and responses occurred in the intervention setting; *A* represents absences; *X* represents peer trainings.

intervals of social initiations and the responses of Eric, Paul, and Chris across all conditions in the intervention setting.

During baseline, the children exhibited few social interactions. Eric's mean numbers of intervals of social behaviors to the trained peer and the generalization peer were 0.67 for total social behaviors, 0.34 for unprompted initiations, and 0.33 for unprompted responses. Paul engaged in no social interactions during this condition, and Chris engaged in responses to the generalization peer with a mean number of intervals of 0.13.

In the initial low-p condition, Eric's social interactions remained low even after the experimenter delivered low-p requests. The mean number of intervals of Eric's total social behaviors to the trained peer and the generalization peer was 3.40. Eric engaged in no unprompted social initiations, although he did engage in unprompted responses to the trained peer with a mean number of intervals of 0.60. The mean number of intervals of Paul's total social behavior was 3.27. He exhibited unprompted responses to the peer with a mean number of intervals of 0.27 with no unprompted social initiations. Chris exhibited low levels of total social behaviors with a mean number of intervals of 0.10 without any unprompted initiations or responses to his trained peer and generalization peer.

During the high-p request sequences with embedded peer modeling, the social interactions among the children increased sharply. Eric's mean numbers of intervals of social behaviors to the trained peer and the generalization peer were 18.15 for total social behaviors, 10.92 for total unprompted initiations and responses, 3.92 for unprompted initiations, and 7.0 for unprompted responses. The data indicate an immediate increase in the level of Eric's interactions with the trained peer and the generalization peer in this condition. Paul's overall social initiations and responses, including unprompted social behaviors, increased from the previous condition. Paul's mean numbers of intervals of social behaviors to the trained peer and the generalization peer were 16.22 for total social behaviors, 8.28 for total unprompted initiations and responses, 0.39 for unprompted initiations, and 7.89 for unprompted responses. Chris's social behavior toward the trained peer and the generalization peer sharply increased in comparison to his behavior in the previous baseline and the low-p request conditions. Chris's mean numbers of intervals of social behaviors to the trained peer and to the generalization peer were 16.16 for total social behaviors, 10.79 for total unprompted initiations and responses, 1.0 for unprompted initiations, and 9.79 for unprompted social responses.

In the second low-p request condition, Eric's mean numbers of intervals of social behaviors to the trained peer and the generalization peer were 26.50 for total social behaviors, 18.75 for total unprompted initiations and responses, 12.75 for unprompted initiations, and 6.01 for unprompted responses. Eric's unprompted social behaviors continued an increasing trend. The mean numbers of intervals of Paul's social behaviors to the trained peer and the generalization peer were 10.18 for total social behaviors, 2.27 for unprompted initiations and responses, 0.45 for initiations, and 1.82 for unprompted responses. Although Paul's unprompted social responses decreased from the previous condition as his trained peer's initiations decreased, his prompted initiations and unprompted initiations maintained the same level as in the previous condition. Chris's unprompted initiations and responses displayed an increasing trend, and in particular, he engaged in more social interactions with the generalization peer during this condition. His mean numbers of intervals of social behaviors to the peer and the generalization peer were 30.14 for total social behaviors, 23 for total unprompted initiations and responses, 4.72 for unprompted initiations, and 18.29 for unprompted responses.

In the maintenance condition, Eric maintained an increased level of unprompted social initiations and responses to the trained peer. His mean numbers of intervals of unprompted social behaviors to the trained peer and to the generalization peer were 23.0 for total unprompted initiations and responses, 14.34 for unprompted initiations, and 8.67 for unprompted responses. Paul's unprompted initiations and responses increased slightly more than in the previous low-p condition, and he exhibited more interactions with the generalization peer. Paul's mean numbers of intervals of social behaviors were 4.75 for total unprompted initiations and responses, 0.50 for unprompted initiations, and 4.25 for unprompted responses. For Chris, one data point was obtained without the generalization peer because of his generalization peer's absence. He engaged in 16.0 intervals of unprompted social behaviors, including four intervals of initiations and 12 intervals of responses to the trained peer.

Social Interactions in the Generalization Setting

The results of each child's number of intervals of unprompted initiations as well as unprompted responses to the trained peer and the generalization peer are represented for all experimental conditions during the generalization sessions. Figure 3 depicts the number of intervals of unprompted social initiations and responses to the trained peer and the generalization peer.

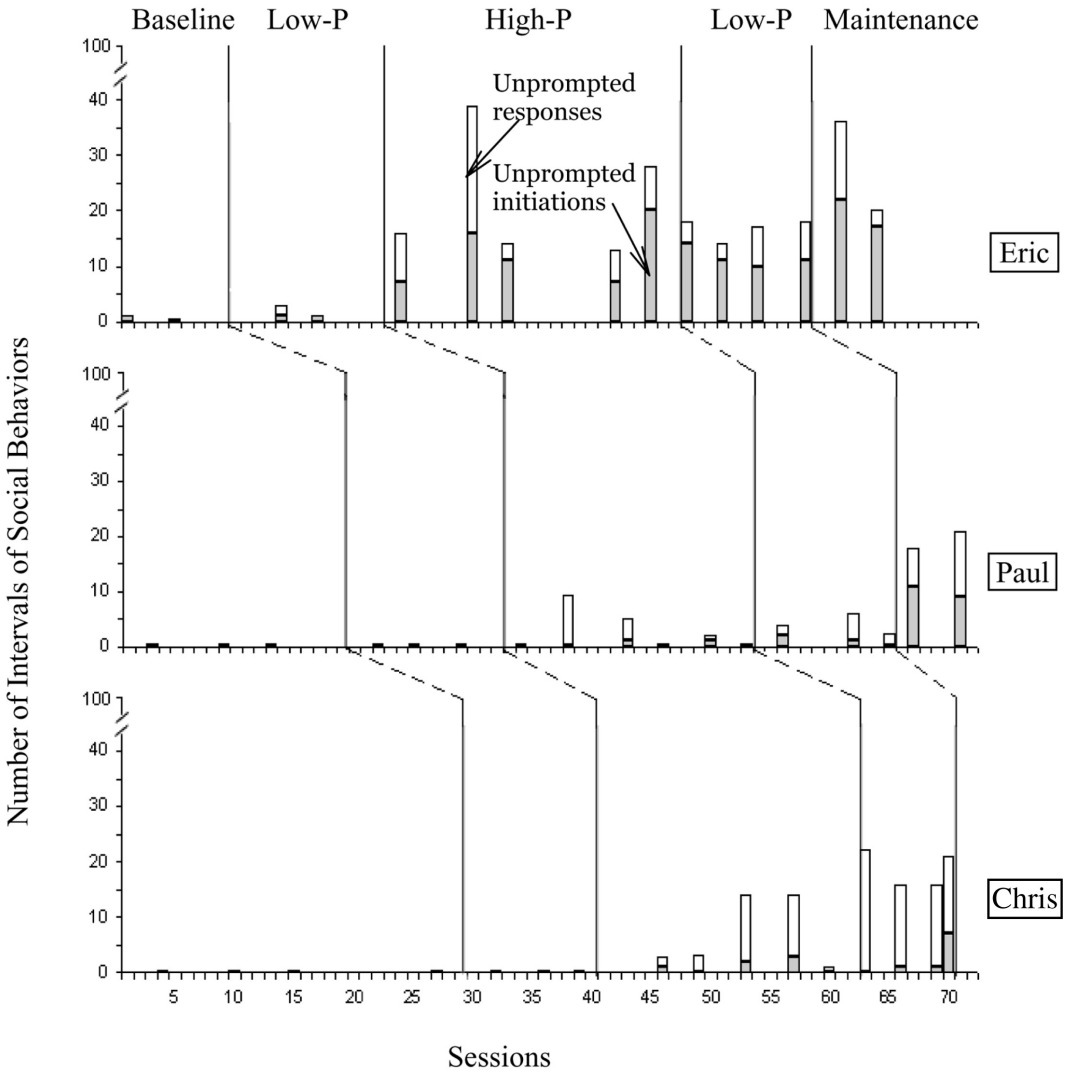
During baseline in the generalization setting, Eric's mean number of intervals of total unprompted social initiations and responses to the trained peer and the generalization peer was 0.50, for which the mean number of intervals of social initiations was 0.50. Paul and Chris engaged in no social interactions with the trained peer or the generalization peer. All target children played by themselves or engaged in stereotypic behaviors, and the trained peers engaged in social interactions mostly with the generalization peers.

In the subsequent initial low-p requests condition in the generalization setting, Eric's interactions slightly increased; however, his interactions remained at low levels. His mean numbers of intervals of unprompted initiations and responses to the trained peer were 1.0 and 1.0, respectively, and he engaged in no social interactions with the generalization peer. Paul and Chris exhibited no interactions with their trained peers or their generalization peers.

During the high-p request sequences with embedded peer modeling in the generalization setting, Eric's mean numbers of intervals of social behaviors to the trained peer and the generalization peer were 22.0 for total unprompted social initiations and responses, 12.20 for unprompted initiations, and 9.8 for unprompted responses. His level of interactions with the generalization peer in the generalization setting as well as with the trained peer sharply and immediately increased after the intervention. Paul's level of interaction also slightly increased while engaging in interactions with the generalization peer. Paul's mean numbers of intervals of social behavior to the trained peer and the generalization peer were 2.67 for unprompted initiations and responses, 0.33 for unprompted initiations, and 2.33 for unprompted responses. Chris's level of initiations and responses increased during this condition in the generalization setting, and he interacted with the generalization peer as well. Chris's mean numbers of intervals of social behaviors to the trained peer and the generalization peer were 7.0 for total unprompted social initiations and responses, 1.20 for unprompted initiations, and 5.80 for unprompted responses.

In the second low-p request during the generalization sessions, Eric's interactions with the trained peer and the generalization peer remained at a high level. His mean numbers of intervals

Figure 3
Number of Intervals of Social Behaviors in the Generalization Setting



of social behaviors to the trained peer and the generalization peer were 16.75 for total unprompted initiations and responses, 13.42 for unprompted initiations, and 5.25 for unprompted responses. Paul's social interactions toward the peer and the generalization peer continued a slight increasing trend during this condition. His mean numbers of intervals of social behaviors to the trained peer and the generalization peer were 4.0 for total unprompted initiations and responses, 1.0 for unprompted initiations, and 3.0 for unprompted responses. Chris's increasing trend in social interactions with the trained peer and the generalization peer continued during the generalization sessions. His mean numbers of intervals of social behaviors to the trained peer and the generalization peer were 18.75 for total unprompted initiations

and responses, 2.25 for unprompted initiations, and 16.50 for unprompted responses. However, he demonstrated no initiations to the generalization peer.

During the generalization sessions in the maintenance condition, Eric's level of social interactions continued to increase. Eric's mean numbers of intervals of social behaviors to the trained peer and to the generalization peer were 28.0 for total unprompted initiations and responses, 19.50 for unprompted initiations, and 8.50 for unprompted responses. Paul's level of interactions with the trained peer and the generalization peer sharply increased. His mean numbers of intervals of social behaviors to the trained peer and the generalization peer were 19.50 for total unprompted initiations and responses, 10.0 for unprompted initiations, and 9.50 for unprompted responses. No sessions were conducted during the maintenance condition in the generalization setting for Chris because of the end of the school year.

Disruptive Behavior

Disruptive behavior was observed to examine if there was a collateral effect in decreasing challenging behaviors as indicated by previous studies (Horner, Day, Sprague, O'Brien, & Healthfield, 1991; Mace & Belfiore, 1990). Eric rarely engaged in disruptive behavior across the experimental conditions in the intervention and the generalization setting (Table 3). However, he exhibited a slight increase in disruptive behavior during the initial low-p request condition. Eric displayed no disruptive behavior during the intervention and the second low-p condition. Paul's disruptive behavior increased with introducing the initial low-p requests, decreased during the high-p request sequences with embedded peer modeling, and remained low during the second low-p condition and maintenance. In the generalization setting, he engaged in few disruptive behaviors across the conditions. Chris's disruptive behavior increased with delivery of the low-p requests and decreased to a low level during the high-p request sequences with embedded peer modeling. His disruptive behavior remained low after the intervention. In the generalization setting, his disruptive behaviors increased during the initial low-p request condition and decreased during the high-p request sequences with embedded peer modeling. Chris engaged in no disruptive behaviors duing the second low-p request condition. For all children, a decrease of a number of general prompts for disruptive behavior was observed once the high-p intervention was implemented (see Table 2).

Social Validity

Regarding the social validity of the intervention goals, three parents of the participants and three early childhood special education teachers at the school indicated that all of the research questions (goals) were useful and valid for children with autism. After viewing videotaped segments from the intervention, 60% of the professionals indicated that the intervention procedure would be easy to use during the student's regular routine, 85% of them responded that they would use the intervention in their classroom, and 85% of them indicated that the intervention was useful to improve social interactions between children.

Table 3
The Percentage of Intervals and Range of the Target Children's Disruptive Behaviors in the Intervention and Generalization Setting

	Intervention Settings					Generalization Settings				
Child	BL	Low-P	High-P	Low-P	Maint.	BL	Low-P	High-P	Low-P	Maint.
Eric										
Mean	0.20	0.40	0	0	0	0	0.50	0	0	0
Range	0-1	0-3	—	—	—	—	0-1	—	—	—
Paul										
Mean	1.60	2.09	0.06	0.58	0	0.33	0.67	0.67	0.67	0
Range	0-10	0-5	0-1	0-1	—	0-1	0-2	0-4	0-1	—
Chris										
Mean	1.13	6.00	0.79	1.43	1.00	0.75	3.33	1.20	0	
Range	0-7	0-14	0-2	0-9	—	0-2	2-6	0-3	—	

Note: BL = baseline; low-p = low-probability request; high-p = high-probability request; maint. = maintenance.

In terms of the intervention outcomes, most of them (95% to 100%) identified that the target children performed the skills listed on the checklist during the intervention condition.

Discussion

The results of the study indicate the responses of the target children to low-p requests to initiate social behaviors to peers increased with the implementation of the high-p requests with embedded peer modeling and were maintained after the removal of the intervention. In addition, the high-p request with embedded peer modeling resulted in increased unprompted social interactions for children. These social interactions were maintained and generalized with the untrained peers and in the nontraining settings. Social validity results indicated a high level of consumer acceptability among parents, teachers, and a group of professionals. The target children's overall social interactions in the intervention and generalization settings met or exceeded the levels of social interactions of the peers without developmental delays observed in the same classroom before intervention.

Our findings support and extend high-p requests and social skill interventions by combining the two strategies. Maintenance and generalization of children's behavior are important concerns for both high-p interventions and social skill interventions. Some researchers using high-p request interventions have shown that compliant responding to low-p requests decreased to baseline levels after withdrawal of the high-p procedure (e.g., Mace et al., 1988; Kennedy, Itkonen, & Lindquist, 1995; Wehby & Hollahan, 2000). Other investigators used fading procedures of the high-p request sequences to maintain the compliant responding to the low-p requests (Ducharme & Worling, 1994; Ardoin et al., 1999). In our study, the compliant responding of the target children was maintained at high levels and continued an increasing trend after removal of the high-p intervention. This outcome has also been demonstrated in previous research (Davis et al., 1992, 1994; Killu et al., 1998).

Although some peer-mediated social skills researchers have reported decreased social interactions after removing teacher prompts (e.g., Odom, Hoyson, Jamieson, & Strain, 1985), the children's social interactions in our study, with the exception of Paul, were maintained and even increased above levels of the previous high-p intervention phase. Although Paul's social behaviors slightly decreased in the intervention setting after removal of the intervention, his social behaviors were maintained at higher levels than during baseline and continued increasing trends in the generalization setting. In addition, the target children's social interactions generalized to the untrained peers in the intervention and generalization setting as demonstrated in the Davis et al. (1994) study.

Several factors may have accounted for these results. In this study, the natural community of reinforcement from children's interactive play and embedding the children's preferences into the requests may have facilitated maintenance of compliant responding to the social requests and social interactions as well as their generalization of the learned social behaviors. In addition, the targeted social requests more likely to facilitate peers' responses may have helped children engage in reciprocal social interactions (Brown & Odom, 1994; Fox & McEvoy, 1993). Additionally, teaching multiple exemplars while implementing the intervention across the play settings may have assisted the target children in maintaining and generalizing their social behaviors (e.g., Goldstein, English, Shafer, & Kaczmarek, 1997).

The peer modeling component may also account for the target children's maintained and generalized social behaviors. Our anecdotal observations suggested the target children exhibited extended play behaviors that the peers demonstrated within a particular play context. As previous investigators have suggested, peer modeling may be able to facilitate untargeted and generalized behaviors in natural environments (e.g., Carr & Darcy, 1990; Venn et al., 1993). We also observed that the peers exhibited more interactive play with the target children after being involved in the high-p procedure as a model. The peers often played with the target children while using or adapting the high-p and low-p requests that the experimenter delivered. They appeared to learn how to play with target children by adapting their play behaviors while participating as models.

Researchers have noted that peer confederates in peer-mediated interventions may become weary of their roles as the initiating playmate for the child with developmental delays; thus, their levels of social initiations might become lower as prompts are removed (e.g., Odom et al., 1987; Odom & Watts, 1991). Odom and Watts (1991) suggested that encouraging reciprocal social interactions of both children with and without developmental delays helps to maintain social interactions better than one-sided interactions that occur when one child is always initiated to by the peer. In our study, peer modeling allowed opportunities for both children to initiate and respond during the intervention as well as to observe play and social behavior. The positive experiences from interactive play may have provided them with mutual reinforcement for engagement in more interactions (Brown & Odom, 1994).

Furthermore, the acceptability and feasibility of the interventions should be carefully considered for implementation in classrooms (Strain & Schwartz, 2001). According to the teachers' reports, the limited available time and resources are primary barriers for implementing social interaction interventions (Odom, McConnell, & Chandler, 1993). However, few previous studies on using the high-p request sequences report social validity (Davis et al., 2000; Ducharme & Worling, 1994; Horner et al., 1991; Killu et al., 1998). The social validity of this intervention was evaluated by parents, teachers, and other early childhood

personnel, who indicated the intervention could be easily and effectively implemented in classrooms.

Implications for Practice

Our study has several implications for practice. Embedding instruction into existing classroom routines is considered to be an effective strategy for use in inclusive settings, as it may provide children with opportunities for practicing social skills in the context of common classroom activities (e.g., Horn, Lieber, Li, Sandall, & Schwartz, 2000; Pretti-Frontczak & Bricker, 2001). In this study, the intervention for the targeted skills was successfully embedded into the children's play during center time activities with peers. Furthermore, the need to use general prompts for children's engagement decreased with the high-p intervention and remained at a low level after removing the intervention. Our results suggest that high-p intervention may be implemented as a less intrusive tactic within ongoing routines while facilitating children's interactive play.

Our data for children's disruptive behavior indicated that those behaviors increased with the implementation of the low-p requests; however, with the introduction of the high-p intervention, disruptive behavior decreased and remained at the same low level even after the withdrawal of the intervention. Often, children with autism tend to engage in disruptive behaviors to avoid demands (e.g., Reese, Richman, Belmont, & Morse, 2005). Researchers from previous studies on high-p interventions have suggested that this intervention strategy may be used to teach demanding tasks while preventing the increase of the children's problem behaviors (Davis, 1995; Davis et al., 1992, 1994, 2000; Davis & Brady, 1993; Davis & Reichle, 1996; Horner et al., 1991).

Our study provided additional support for the use of peers without developmental delays as effective intervention agents for observational learning and facilitating positive social relationships among children. Young peers provide opportunities for children's engagement in social interactions in ways that adults cannot (Erwin & Schreiber, 1999) and have been observed to make accommodations appropriate to the individual needs of children with developmental delays during play (Guralnick, 1999). A structured peer modeling procedure incorporating the high-p request sequence may be an effective and efficient tactic for teachers to use in inclusive settings.

Future Directions for Research

Our findings raise additional research questions relative to the utility of the high-p and social skill interventions. Although researchers using high-p request sequences have demonstrated its effectiveness on facilitating functional skills as well as compliance, only a few investigations have been conducted with young children (Banda, Neisworth, & Lee, 2003). In addition, few studies have incorporated measures of social validity with regard to high-p interventions. Additional replications are needed to better investigate the effectiveness of the high-p interventions in teaching young children with developmental delays and to assess more broadly the feasibility and acceptability of the procedure in early childhood and early childhood special education classrooms.

In our study, the high-p request sequences with embedded peer modeling were delivered as a multicomponent intervention package; therefore, the effectiveness of each intervention

component on compliant responding to requests and social behaviors was not systematically examined. In the future, researchers might investigate how various components of the combined interventions might affect children's social behavior and identify more efficient interventions to increase the targeted social skills in young children with autism.

Researchers may also want to carefully investigate the effectiveness of the social interaction interventions on children with autism who are nonverbal or have limited or extremely limited language skills. Investigators have suggested that future research on social behavior should attempt to determine the differential effects of interventions and settings on children with different characteristics (e.g., Guralnick, 1997; Odom, 2002). Social skills are complex and require the development of other related skills, such as communication, cognitive, play, and motor skills (Bailey & Wolery, 1992; Guralnick, 1990). In particular, children with little or no productive language rarely initiate interactions and engage mostly in solitary behavior (Harper & McCluskey, 2002). In this study, the 3 target children's social interactions increased with the high-p intervention; nevertheless, each child demonstrated different levels of social interaction. The first target child, Eric, showed a sharp increase in interactions, and he engaged in more initiations than responses in both the intervention and the generalization settings. The social behavior of the other children focused on responses to their peers, although their social initiations increased with the implementation of the intervention and showed increasing trends in the generalization setting. Children's differing communication levels may have influenced their ability to respond to the intervention and their social outcomes. As the target children and the peers engaged in the interactive play, we observed that they indicated mutual enjoyment while focusing most often on manipulative play. In the future, researchers may examine the combined strategy of high-p requests with embedded peer modeling in the context of preferred games and activities adapted to children's communication skills and preferences for play.

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